

German Israel Gonzalez Roman

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OBJECTIVE

Electrical Engineering student seeking internship/co-op to apply technical skills, gain practical experience, and contribute to engineering projects in a professional environment.

EDUCATION

University of South Florida

Tampa, FL

Bachelor of Science in Electrical Engineering • USF GPA: 3.98

Expected December 2026

Relevant Coursework: Signals and Systems, Electronics I, Semiconductor Devices, Power Electronics, Linear Control Systems, Embedded Systems, Microprocessors Systems, Electromechanical Systems.

SKILLS

- Electrical: AC/DC circuit design and analysis, control systems, power circuits and transformers, harmonic analysis, PID design, PLC programming, Ladder Logic, electrical schematic interpretation, system modeling and simulation.
- Programming C, Python, Assembly, VHDL (FPGA – Xilinx), ROS2 (Ubuntu, Raspberry Pi 5), RAPID (ABB), ACE 4.7 (Omron Adept) motion programming, digital logic design.
- Tools & hardware: MATLAB, Simulink, LabVIEW, RS Logix 5000, CCW, Fusion 360, RobotStudio, oscilloscope, multimeter, power supplies, soldering and prototyping, circuit debugging, 3D modeling, Microsoft Office.

WORK EXPERIENCE

University of South Florida – Tampa, FL

September 2025 – Present

Student Assistant – Automation & Robotics Laboratory

- Selected by the Department Chair of Industrial Engineering to develop control programs and deliver technical demonstrations for automation laboratory systems to sponsors and university audiences.
- Support 60+ students per week in PLC programming and the operation of automated SCARA, 6-DOF, and Delta platforms, ensuring successful completion of laboratory objectives.

University of South Florida – Tampa, FL

May 2025 – September 2025

Undergraduate Researcher – Automation & Robotics Laboratory

- Designed and modeled a delta robot structural frame in Fusion 360 per manufacturer specifications; performed mechanical assembly and integrated power, control, and communication subsystems supporting calibration, commissioning, end-effector integration, and ACE motion programming.
- Presented the completed robotic system to the Industrial Engineering Department Chair, securing approval for departmental integration and developing supporting instructional materials.

United Air Conditioning – Largo, FL

March 2019 – June 2022

HVAC Technician – Service & Installation

- Diagnosed and resolved electrical and mechanical faults across 80+ residential and commercial systems weekly using electrical schematics and wiring diagrams, implementing corrective actions including component replacement, electrical modifications, and system functional verification.
- Promoted to installation team lead, planning system installations and making on-site technical decisions to address constraints, resolve commissioning issues, and support operational efficiency.

Polytechnic Institute of Tampa – Clearwater, FL

August 2019 – July 2021

Instructor

- Taught electricity and HVAC courses, defining course structure, content, and laboratory exercises.

Adecco – Clearwater, FL

February 2018 – January 2019

Contract Employee

- Assembled primary and secondary windings for transformers at ITI (General Electric).

HONORS

- Member, Tau Beta Pi Engineering Honor Society, November 2024– Present.
- Dean's List, Spring 2024 – Present.